

Operating Systems 305

Group Project

Task 1

Part I: Counting System Calls

You are required to implement the project details specified in:

<https://github.com/remzi-arpacidusseau/ostep-projects/tree/master/initial-xv6>

To check that your implementation is correct: run the tests provided with the project. [You will need to setup the test files before testing your program].

Part II: System Call Tracing

Modify the xv6 kernel to print out a line for each system call invocation. It is enough to print the name of the system call and the return value; you don't need to print the system call arguments.

When you're done, you should see output similar to this when booting xv6:

```
...
fork -> 2
exec -> 0
open -> 3
close -> 0
$write -> 1
write -> 1
```

That is init forking and execing sh, sh making sure only two file descriptors are open, and sh writing the \$ prompt. (Note: the output of the shell and the system call trace are intermixed, because the shell uses the write syscall to print its output.)

Hint: modify the syscall() function in syscall.c.

Optional challenge: print the system call arguments.